

EMPLOYMENT

Freight Desk Developer	Mercuria Energy Trading London, UK	May 2025 – present
- Built a freight exposure and P&L attribution engine, decomposing daily P&L across hundreds of thousands of voyages, FFAs, physical fixtures, and chartering events to improve front-office visibility into desk performance and risk. - Architected a Dagster-based CDC pipeline streaming trade and position data from multiple ETRMs into an Iceberg data lake, reducing data latency from 15+ minutes to under 1 minute and improving SLA consistency across downstream consumers. - Partnered with traders, risk, and middle office stakeholders to productionize risk and P&L workflows, consolidating fragmented manual processes into a single deployable reporting stack. - Developed internal AI tooling and developer guidance, including Grafana MCP integrations and adoption playbooks for agentic coding workflows, to accelerate engineering productivity across the team.		
Front Office IT Intern	Castleton Commodities Houston, TX	June 2024 – August 2024
- Built a full-stack pipeline visualization geospatial platform using Next.js, Snowflake, Python, AWS, and Mapbox, giving traders real-time visibility into US natural gas flows, nominations, and positions. - Engineered a graph-based model of a 100,000+-mile US pipeline network using H3 geospatial indexing, reducing nomination routing time by 40%. - Designed a Snowflake schema with optimized tables supporting sub-second queries for 15+ concurrent trader sessions during peak trading hours. - Delivered a production trading tool adopted by the Houston gas desk for daily workflow management.		
Teaching Aide EECS 280	UM CSE Faculty Ann Arbor, MI	January 2022 – December 2023
- Led weekly lab sections for 40 students in a core programming course with 3,000+ annual enrollments. - Supported inclusive teaching initiatives and student community-building events with faculty and staff.		

PROJECTS

Ekko: AI Language Learning Assistant	2026
- Developed a mobile-first language learning platform using Expo, Supabase, and AWS, with a context-aware LLM interface and persistent user memory. - Implemented retrieval and state persistence mechanisms to track user mistakes, vocabulary retention, and long-term learning progress across sessions. - Architected the system to support adaptive, context-rich tutoring workflows that improve relevance and continuity over time.	

RESEARCH

Graduate Thesis	University of Michigan	2026
Stochastic Techno-Economic Analysis of Hydrogen Storage Using Reinforcement Learning		
- Built an innovative valuation framework for hydrogen storage using reinforcement learning, Monte Carlo simulation, and LP benchmarks across four storage pathways – LH₂, LNH₃, LOHC, and methanol. - Formulated hydrogen storage dispatch as a constrained sequential optimization problem and trained Maskable PPO agents in a custom Gymnasium environment with pathway-specific physical and operating constraints. - Developed a stochastic forward-curve simulation pipeline using ARMA-GARCH-generated shape factors and basis-function decomposition to reconstruct synthetic hydrogen price term structures from TTF dynamics. - Found that LNH₃ showed the strongest storage potential due to its low operating friction and favorable long-term economics.		

EDUCATION

University of Michigan	Ann Arbor, MI	September 2024 – April 2026
M.S in Data Science, GPA: 3.97 Graduate Coursework: Deep Learning, Generative AI, Unsupervised Learning, Supervised Learning, Machine Learning Pipelines		
University of Michigan	Ann Arbor, MI	September 2021 – April 2024
B.S.E in Computer Science, GPA: 3.78 Undergraduate Coursework: Thermodynamics, Organic Chemistry, Linear Algebra, Data Structures, Databases, Cybersecurity		

AWARDS AND CERTIFICATIONS

- Mercuria USA Energy Hackathon 2023:** 2nd Place Overall
- Canadian Hydrogen Convention 2025:** Grand Prize Winner – Global Hydrogen Case Competition
- Amazon Web Services** Certified Solutions Architect – Associate

SKILLS

- C++, Python, SQL, Pandas, Streamlit, Tensorflow, Excel, AWS, Altair, React, Dagster, Claude, Docker, Spanish (B2), Chinese (C1)